

classes that have low positive correlation do have some diversification benefit, especially if you hold several types in a portfolio.

THE TWO-ASSET-CLASS MODEL

Finance professors begin teaching asset allocation techniques using two asset classes. The students learn about correlation, risk reduction, and the efficient frontier in a simple model of two investments that have low correlation with each other. After the students have mastered an understanding of the benefits of asset allocation using two investments, the professor expands the exercise into a multi-asset portfolio by adding a third, fourth, fifth, and sixth investment category. The remainder of this chapter follows the same path by explaining asset allocation using a two-asset-class portfolio consisting of U.S. stocks and U.S. Treasury bonds. [Chapter 4](#) expands the discussion into a multi-asset-class model.

The two asset classes examined in this chapter are a U.S. large-stock index and an intermediate-term Treasury-note index. The S&P 500, an index of 500 leading U.S. corporations, is used as a proxy for U.S. large-stock returns. The Treasury note returns are based on two data series. Prior to 1973, the Treasury return is represented by the return on five-year Treasury bonds. Starting in 1973, the Treasury note returns represent the performance of the Barclays 1–10 Year Treasury Index, which is a diversified portfolio of short- to intermediate-term U.S. Treasury securities.

RISK-AND-RETURN FIGURES

In this chapter and for the remainder of the book, portfolio risk and return are illustrated using charts and tables. [Figure 3-6](#) represents a classic risk-and-return frontier. The vertical axis in [Figure 3-6](#) represents the compounded annualized return of a series of portfolios, and the horizontal axis is the risk as measured by the standard deviation of those annual returns.

FIGURE 3-6 Classic Risk-and-Return Efficient Frontier